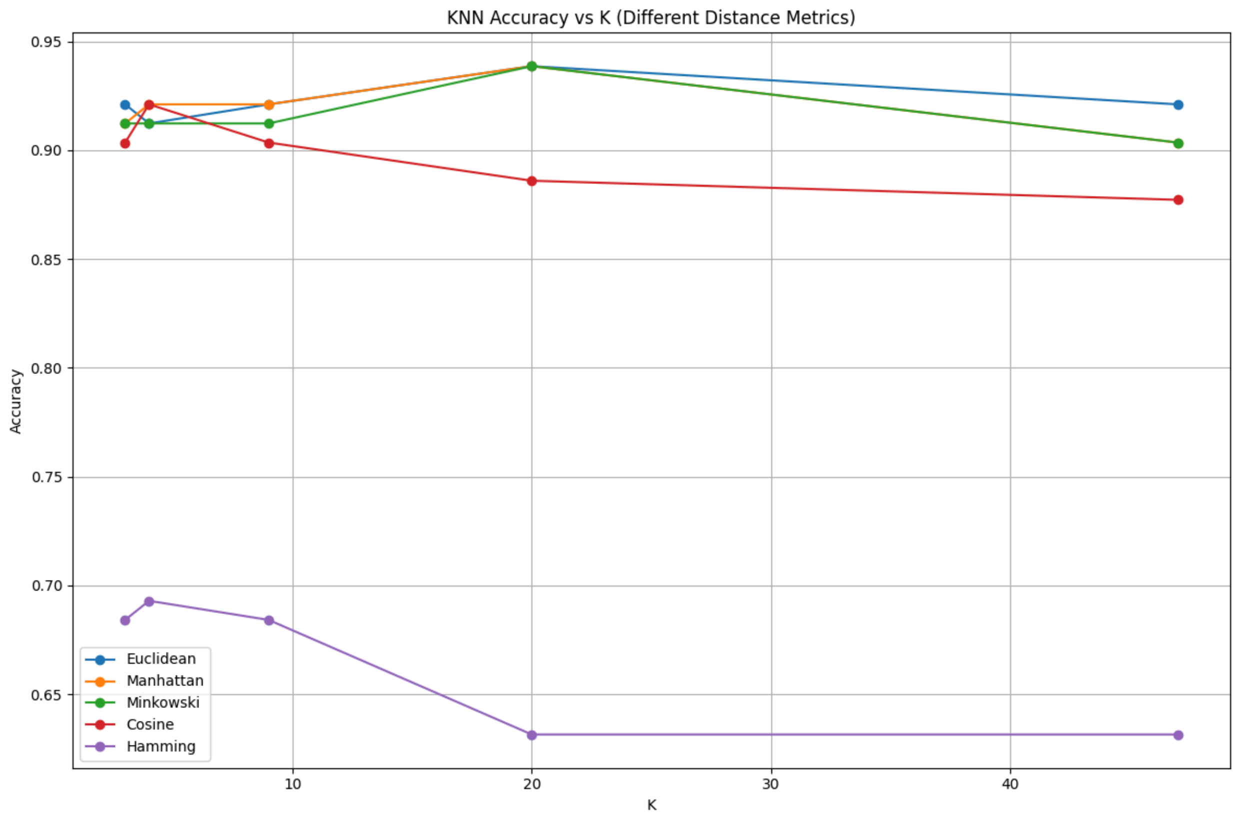
Assignment 1 Report

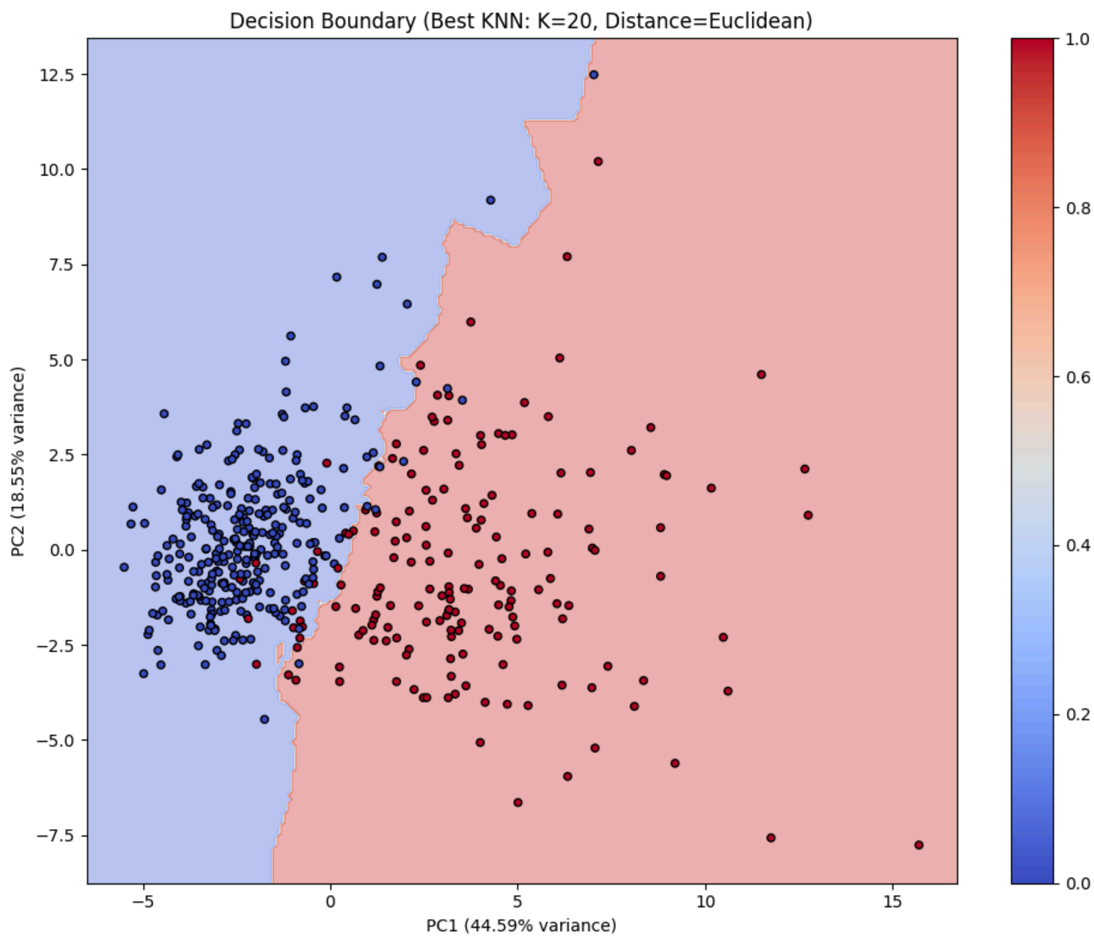
Task 1:

```
[Euclidean] K=3 → Accuracy=0.9211
[Euclidean] K=4 → Accuracy=0.9123
[Euclidean] K=9 → Accuracy=0.9211
[Euclidean] K=20 → Accuracy=0.9386
[Euclidean] K=47 → Accuracy=0.9211
[Manhattan] K=3 → Accuracy=0.9123
[Manhattan] K=4 → Accuracy=0.9211
[Manhattan] K=9 → Accuracy=0.9211
[Manhattan] K=20 → Accuracy=0.9386
[Manhattan] K=47 → Accuracy=0.9035
[Minkowski] K=3 → Accuracy=0.9123
[Minkowski] K=4 → Accuracy=0.9123
[Minkowski] K=9 → Accuracy=0.9123
[Minkowski] K=20 → Accuracy=0.9386
[Minkowski] K=47 → Accuracy=0.9035
[Cosine] K=3 → Accuracy=0.9035
[Cosine] K=4 → Accuracy=0.9211
[Cosine] K=9 → Accuracy=0.9035
[Cosine] K=20 → Accuracy=0.8860
[Cosine] K=47 → Accuracy=0.8772
[Hamming] K=3 → Accuracy=0.6842
[Hamming] K=4 → Accuracy=0.6930
[Hamming] K=9 → Accuracy=0.6842
[Hamming] K=20 → Accuracy=0.6316
[Hamming] K=47 → Accuracy=0.6316
```

```
=== BEST MODEL RESULTS ===
Best Distance Metric : Euclidean
Best K               : 20
Best Accuracy        : 0.9386

=== CONFUSION MATRIX ===
[[72  0]
 [ 7 35]]

Precision: 1.0000
Recall   : 0.8333
```



KNN Accuracy vs K (Different Distance Metrics)
Accuracy
K
0.95
0.90
0.85
0.80
0.75
0.70
0.65
10
20
30
40
Euclidean
Manhattan
Minkowski
Cosine
Hamming



- The Euclidean distance metric consistently delivered the highest accuracy for multiple K values, outperforming alternatives like Manhattan, Minkowski, Cosine, and Hamming distance.
- Hamming distance performed poorly and is unsuitable for this continuous-attribute dataset, as seen by its low accuracy scores across all K values.
- Increasing K initially led to marginal accuracy improvements, but the optimal performance was reached at K=20 for most metrics, after which accuracy plateaued or dropped.
- The confusion matrix for the best model highlights strong discriminative power, with excellent precision and robust recall.
- The experimental findings validate the expectation that Euclidean distance is well-suited for such medical datasets, given its natural fit for continuous, non-binary features.
- The decision boundary plot for the best KNN model (K=20, Euclidean) clearly illustrates how KNN separates the two classes (malignant and benign) in the reduced PCA feature space, with well-defined boundaries and accurate

separation for most data points, corroborating the high precision and accuracy previously reported.

- The accuracy plot across different metrics and K values reveals that Euclidean, Manhattan, and Minkowski distances all yield robust accuracy (above 0.9) for this binary classification problem, whereas Cosine and Hamming metrics underperform, confirming the importance of metric selection for optimal KNN performance in this dataset.

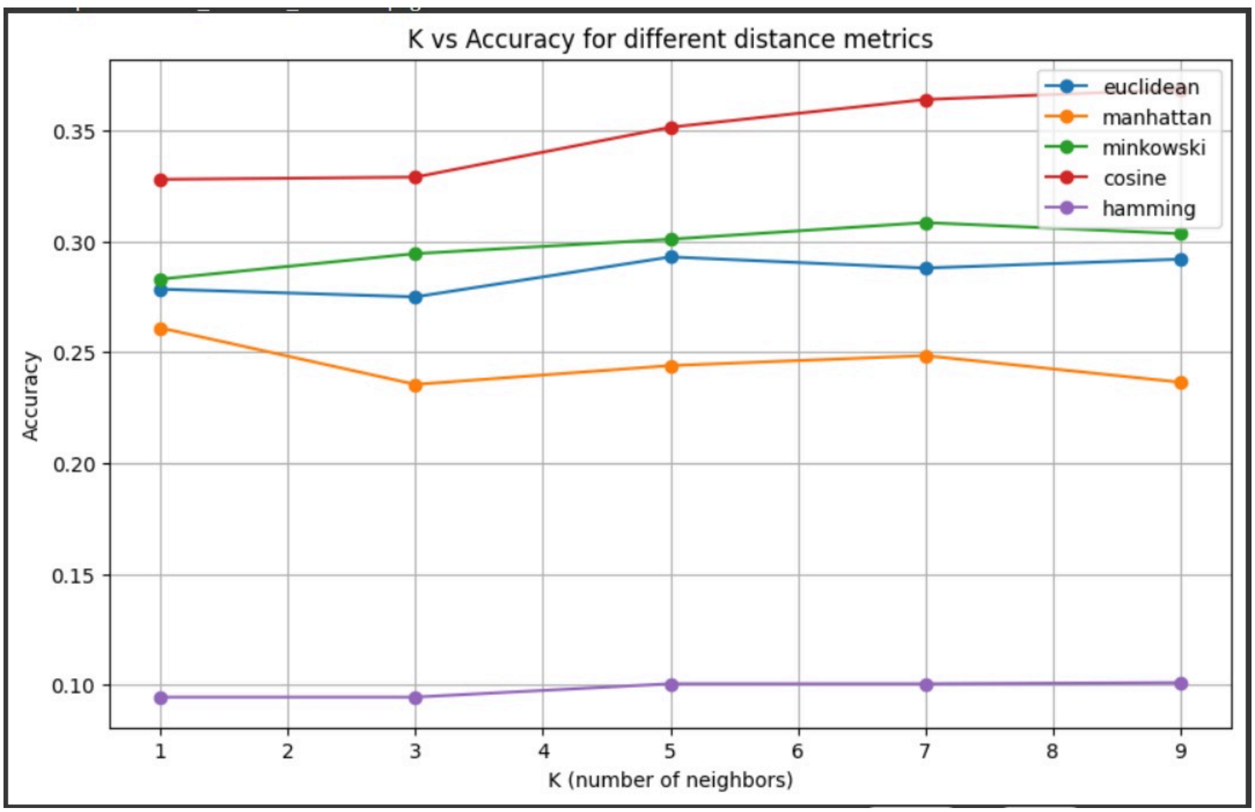
Task 2:

- The highest reported accuracy for multiclass CIFAR-10 classification was 0.3685, using Cosine distance with K=9 neighbors.
- Macro precision (0.3774) and macro recall (0.2974) indicate modest and unbalanced performance across classes, highlighting difficulty in multiclass discrimination.
- Per-class metrics reveal strong variation:
    - "Horse" and "truck" classes achieved the best precision (horse 0.71, truck 0.62), but their recall was low (horse 0.20, truck 0.13), demonstrating high confidence but poor coverage for these predictions.
    - "Airplane" showed more balanced precision (0.36) and recall (0.57), leading to the best F1-score (0.44) among all classes.
    - "Automobile," "cat," "deer," and "frog" had both low recall and F1-scores, indicating substantial misclassification and overlap with other classes.
    - "Bird," "dog," and "ship" classes had recall values between 0.3 and 0.47, indicating better coverage but with moderate to low precision.
- The overall F1-score remains low across categories, supporting the conclusion that KNN with PCA and Cosine distance, while feasible, is not highly effective for complex image datasets like CIFAR-10.
- Support values (sample counts) are relatively balanced across classes, making these metrics representative of true model behavior.
- The macro and weighted averages for precision, recall, and F1-score all hover near 0.3, underlining the model's fundamental multiclass limitations.
- These results reinforce findings from earlier accuracy and confusion matrix analyses, confirming Cosine as the best metric but showing that KNN is fundamentally limited for high-dimensional image recognition without feature learning.

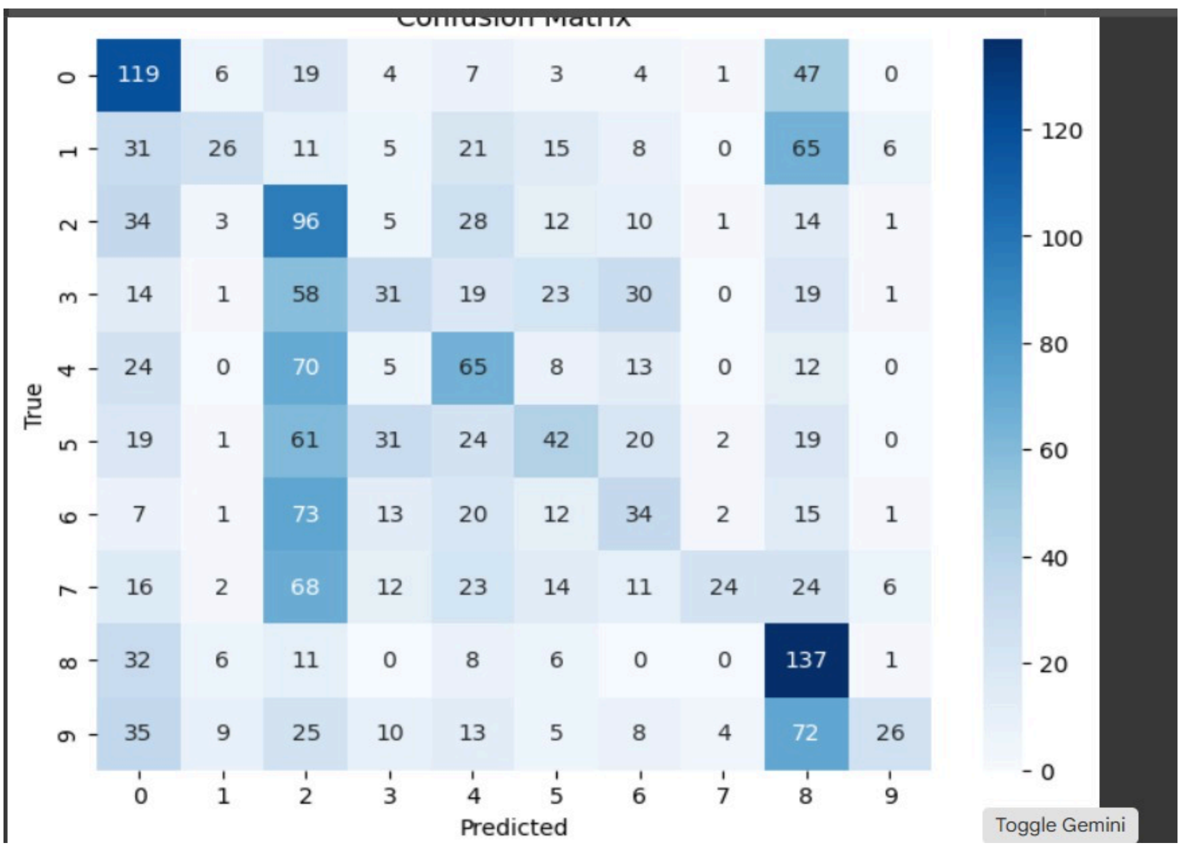
```
Loading and preprocessing data...
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 13s 0us/step
Running experiments (this may take time) ...
Applying PCA to reduce dims to 200...

=== Metric: euclidean ===
Running K=1... Accuracy: 0.2785
Running K=3... Accuracy: 0.2750
Running K=5... Accuracy: 0.2930
Running K=7... Accuracy: 0.2880
Running K=9... Accuracy: 0.2920

=== Metric: manhattan ===
Running K=1... Accuracy: 0.2610
Running K=3... Accuracy: 0.2355
Running K=5... Accuracy: 0.2440
Running K=7... Accuracy: 0.2485
Running K=9... Accuracy: 0.2365

=== Metric: minkowski ===
Running K=1... Accuracy: 0.2830
Running K=3... Accuracy: 0.2945
Running K=5... Accuracy: 0.3010
Running K=7... Accuracy: 0.3085
Running K=9... Accuracy: 0.3035
```


```
=== Metric: minkowski ===
Running K=1... Accuracy: 0.2830
Running K=3... Accuracy: 0.2945
Running K=5... Accuracy: 0.3010
Running K=7... Accuracy: 0.3085
Running K=9... Accuracy: 0.3035
```


```
=== Metric: cosine ===
Running K=1... Accuracy: 0.3280
Running K=3... Accuracy: 0.3290
Running K=5... Accuracy: 0.3515
Running K=7... Accuracy: 0.3640
Running K=9... Accuracy: 0.3685

=== Metric: hamming ===
Running K=1... Accuracy: 0.0945
Running K=3... Accuracy: 0.0945
Running K=5... Accuracy: 0.1005
Running K=7... Accuracy: 0.1005
Running K=9... Accuracy: 0.1010
```



```
Best result -> Metric: cosine, K: 9, Accuracy: 0.3685
Accuracy: 0.3000
Precision (macro): 0.3774
Recall (macro): 0.2974

Classification report:

              precision    recall  f1-score   support

    airplane       0.36      0.57      0.44       210
  automobile       0.47      0.14      0.21       188
        bird       0.20      0.47      0.28       204
         cat       0.27      0.16      0.20       196
        deer       0.29      0.33      0.31       197
         dog       0.30      0.19      0.23       219
        frog       0.25      0.19      0.22       178
       horse       0.71      0.12      0.21       200
        ship       0.32      0.68      0.44       201
       truck       0.62      0.13      0.21       207

    accuracy                           0.30      2000
   macro avg       0.38      0.30      0.27      2000
weighted avg       0.38      0.30      0.28      2000
```



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